

## STAT301 In-class Activity 9

Medical professionals often use the CD4 count to decide when to begin treatment during HIV infection. There is an HIV dataset on the class website. We wish to model CD4 count, with treatment arms (“trarm”), viral load (“logRNA”, in  $\log_{10}$ -scale), age, weight as potential predictors.

(a) Since CD4 is a count, we might consider a Poisson GLM, with CD4 as a response and all predictors in the model. Is there a significant difference in CD4 counts between the treatment arms (adjusted for covariates)?

(b) Is there an overdispersion problem for the above Poisson model?